Final project report and presentation

The final deliverable for the course project is a written report and an in-class presentation on May 27. Your objective for the final deliverable is to **conduct new work that incorporates network analysis** into your analysis of the world models, and to **refine and integrate your work from the milestone**. Your ultimate goal is to define the set of parameters that best enables sustainable development as you define it.

Incorporating network analysis

One of the challenges with the world3 model is that due to its complexity, it is difficult to visualize the components of the model and gain an understanding of the overall model structure. To help us better understand this complex model, we will translate the systems dynamics model into a network.

Network analysis as a tool has many potential use cases, as we discuss in the course. Analyzing the structure of system dynamics models is a very specific application of network analysis, but nevertheless it enables us to gain an appreciation for the capabilities of network analysis.

Loading the world3 model as a network

Your first step is to load the world3 model structure as a directed network using networkx. Nodes represent the different variables in the model (stocks, flows, and parameters/variables), and edges represent the influence from one variable to another.

You may use the following code to load your network. The variables of the model are listed in the "world3-03_variables.json" file, available on Moodle. The code reads the file and creates a directed network, where an edge originates at a certain variable and ends at the variable that is influenced by the origin variable.

```
import json
import networkx as nx

json_file = open('world3-03_variables.json')
w3_vars = json.loads(json_file.read())
json_file.close()

G = nx.DiGraph()
for name, val in w3_vars.items():
    G.add_node(name, var_type=val['type'])
    if val['dependencies'] != None:
        G.add_edges_from([(dep, name) for dep in val['dependencies']])
```

Note that when the nodes are added, they are added along with a variable called "`var_type`." The nodes that represent stocks have the value "`stateful`" as their `var_type`.

Analyzing the network

Now that you have your network loaded, use the tools discussed in class (and any other network analysis tools that you learn about on your own) to analyze the network. Here are some questions to get you started but be sure to go beyond these questions in your project.

- Which variables are considered "important" in the network? How does your specific measure of "importance" affect this consideration?
- Can you identify any cycles? What do these cycles represent? What about the length of the cycles?
- Are there any natural communities in the network structure?

Note that some network analysis tools require that the network be undirected rather than directed. In networkx it is possible to translate a directed graph to an undirected one.

Implications for system dynamics policy development

Considering your network analysis of the system dynamics model, think about how the policy development you proposed in the first milestone could be updated or informed by your findings. Include a discussion of this, with a view toward your final project presentation.

Resources

networkx documentation

Final report

The assessment of the report is based on the rubric shown below. The report is maximum 10 pages and covers the work throughout the whole semester. You must also submit a final compilation of your group's code on Moodle.

Criteria	Score
Definition of sustainable development metric Clarity of the definition: be explicit in how you construct your metric(s). Rational for the definition: defend your choices.	5
Evidence of network analysis Centrality, cycles, communities, and additional tools. How does the network analysis impact your policy development?	15
Evidence of system dynamics analysis Feedback loops, parameter effects, and additional analysis.	15
Automated policy design Quality of the automated process designed to evaluate different policy scenarios and the presentation of the results. How does the model perform in the end? (including analysis from both systems dynamics and network analysis parts)	15

Discussion of results and implications What do your findings suggest? What had the biggest impact in moving your model toward your sustainable development target? What are the biggest strengths and weaknesses of both the model and your approach?	15
Overall creativity Evidence of going beyond the bare minimum by exploring additional parameters, novel ideas, custom visualizations, or advanced methods.	15
Presentation quality Clear and organized presentation of the analysis and results, including visualizations.	10
Code submission Does it contain all the code you reference in your final presentation? Is it well-documented (i.e. commented)?	10

Presentation

The presentation is 10 minutes followed by Q+A. The presentation should follow the same structure as the final report and will be assessed with the same rubric framework (except for the code submission part).